



LAR Heater Pigging & Decoking Services

Marathon Petroleum, Los Angeles Refinery

Carson Coker #1 & #2, Wilmington DCU H-101

Presented by Marshall Douglas
Director of Pigging Operations
DSP#26098 | September 2026



Agenda

Structured to the evaluation team's ten discussion points



- | | | | |
|-----------|-----------------------------------|-----------|---|
| 01 | Pigging & decoking methodology | 06 | Emergency response & local resources |
| 02 | Execution plan & schedule | 07 | Safety & risk mitigation |
| 03 | Operational readiness & resources | 08 | Best practices, capabilities & innovation |
| 04 | Equipment & technology | 09 | MPC references & case studies |
| 05 | Cleanliness verification & QA/QC | 10 | Why USA DeBusk |

Our understanding of the scope

Recurring mechanical pigging & decoking, lump sum per occurrence, mobilization and demobilization waived

Unit	Heater	Passes	Per year	5 years
Carson Coker #1	Heater #1	2	4	20
Carson Coker #1	Heater #2	2	4	20
Carson Coker #2	Heater #3	2	3	15
Wilmington DCU	H-101	4	1	5

Source: DSP#26098 Rev 1, scope table

4

heaters, two sites

12

decokes per year

60

decokes over 5 years

2,790

project hours, 5 years



Marathon Los Angeles Refinery
Carson delayed coker

Photo: Marathon Petroleum

Bi-directional mechanical decoking

Every pass follows the same controlled sequence

1

Rig-in & pressure test

Connect 300# launchers and receivers; fill and test before any launch

2

Baseline flow test

Record pressure at a set rate for a before-and-after comparison

3

Gauge foam run

Proves the pass is open and sets the initial pig size

4

Scraper pig progression

Carbide-studded pigs run in both directions, stepped up $\frac{1}{8}$ " per clean pass

5

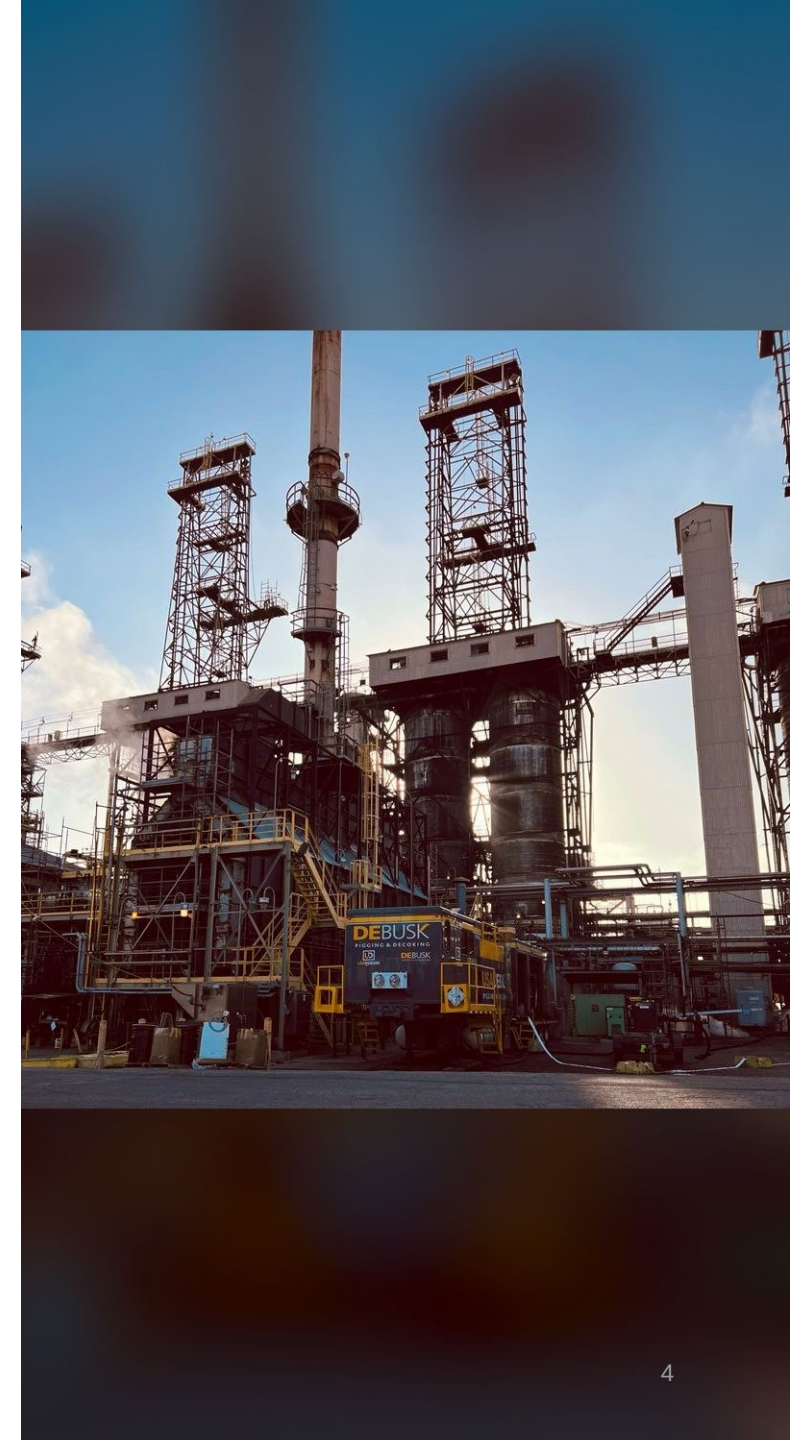
Oversize polish

Up to tube ID + $\frac{1}{4}$ " maximum, with real-time pressure monitoring

6

Gauge foam & sign-off

Line-size foam witnessed by Marathon, final flow test, pass signed clean



Pig progression plan by heater

Answers RFI item SP-2; the gauge foam run sets the starting size, the tube ID sets the ceiling

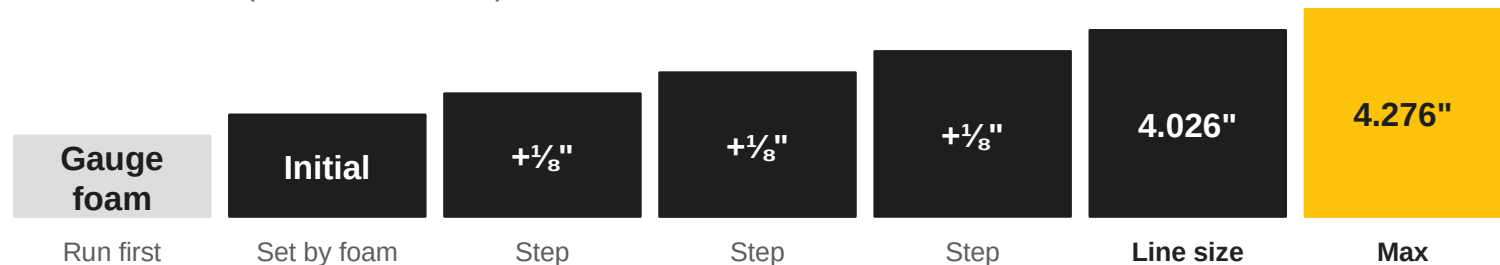
Carson Cokers, Heaters #1, #2, #3

Governing ID 3.826" (4" Sch 80)



Wilmington DCU H-101

Tube ID 4.026" (4.5" OD × 0.237")



The sizing rule

Max pig OD =
smallest tube ID + 0.250"

- Step up 1/8" after each clean pass
- Pig wear drives size & appendages
- Body length set for bends & mule ears
- Gauge foam sets the starting size
- Line-size foam witnessed at sign-off

Engineered decoking pigs, matched to the coke

The most diverse selection of engineered decoking pigs in the world



Foam, carbide-studded and worn pigs from USA DeBusk jobs



Urethane body, foam core

Flexes through return bends while flow bypasses the pig and carries debris away



Durometer chosen per job

Softer compounds compress through tight bends; harder compounds remove coke aggressively



Bald or studded

Carbide studs as standard, with alternate stud materials matched to the tube metallurgy



Heavy-coke designs

High-durometer, fully studded pigs brought in when pig wear shows hard, heavy coke

Heater data & coil-specific controls

Heater	Section	Passes	Footage	Tube ID	Metallurgy
Carson Cokers	Convection	2	2,268'	3.826"	SS 321
Carson Cokers	Radiant	2	4,032'	3.826" / 4.026"	A335 P9
H-101	Convection	4	2,508'	4.026"	A312 347H
H-101	Radiant	4	7,524'	4.026"	A312 347H

Source: DSP#26098 Rev 1, Technical Data (coil data provided by Marathon)

6,300'

per Carson heater

10,032'

on H-101

Grade

launcher access (H-101
radiant: grade & 1st deck)



Stainless coils

Water quality verified before fill;
passivation coordinated with Marathon; no
galvanized contact with tubes



Connections

600# RFWN coil flanges adapted to USA
DeBusk 300# launchers and receivers

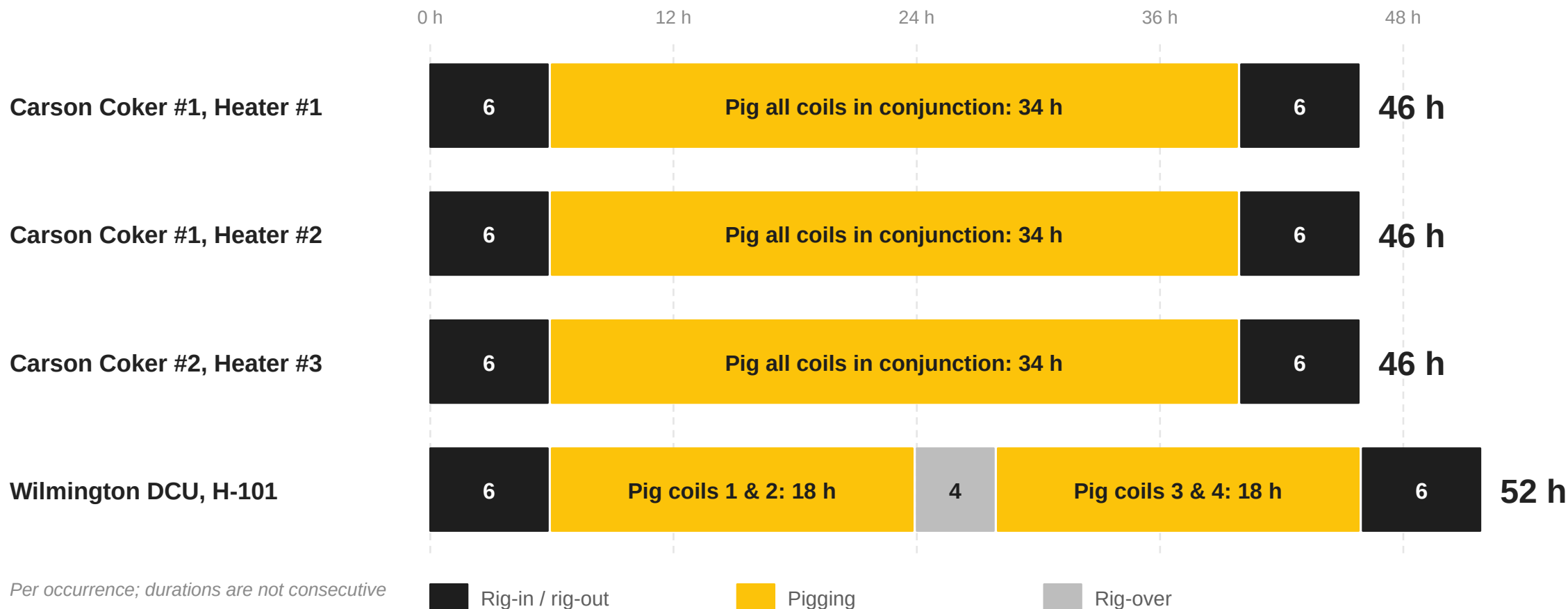


Return bends

Pig length and flexibility selected for 180°
returns and mule ears

Execution plan per occurrence

Continuous 24/7 pigging on 12-hour shifts; passes pigged in conjunction



Five-year program at a glance

Illustrative annual cadence: dates are set to Marathon's run-length and turnaround plan



12 decokes a year

60 decokes in 5 years

2,790 project hours

\$0 mob / demob

Same crew leadership, same equipment and same procedures every visit: each decoke builds on the last one's records.

Crew & equipment, every occurrence



Equipment

- 1 TriMax triple pumper
- 1 Support unit: fittings, tooling, spares
- 2 Crew trucks



Manpower

- 2 Supervisors: 1 day, 1 night
- 3 Operators: 2 day, 1 night
- 1 Project manager

~5%

crew turnover, last 12 months

24/7

two 12-hour shifts

100%

training verified before mobilization

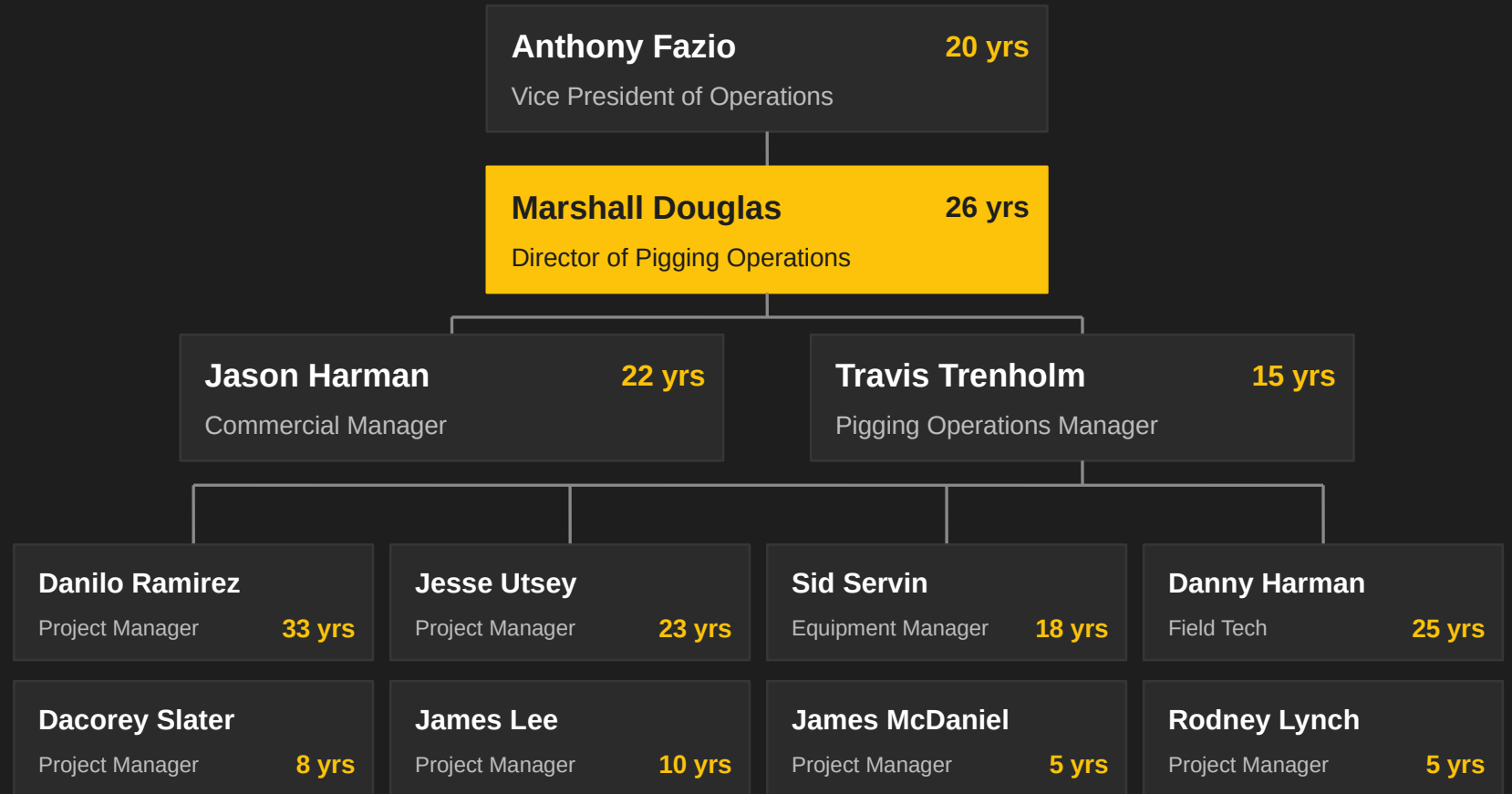
Your LAR project team

210

years of combined pigging & decoking experience

15+

years self-performing heater pigging



The TriMax triple pumper

6 identical TriMax units, 18 engines: one standardized fleet



Three independent pump systems

Closed-loop, bi-directional; 50% more productive than a twin pumper



Automated HMI control

Three 17" touchscreens, automated digital pig logs



150–300 psi working pressure

Pump rated 600 psi; 500–550 psi effective after valve losses



Real-time sensor data

Coke shows as pressure spikes at the operator interface



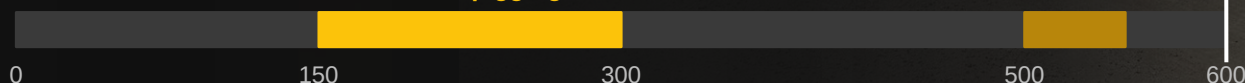
Cummins QSL9, 333 bhp, Tier 4 Final

CARB-registered engines, California-ready

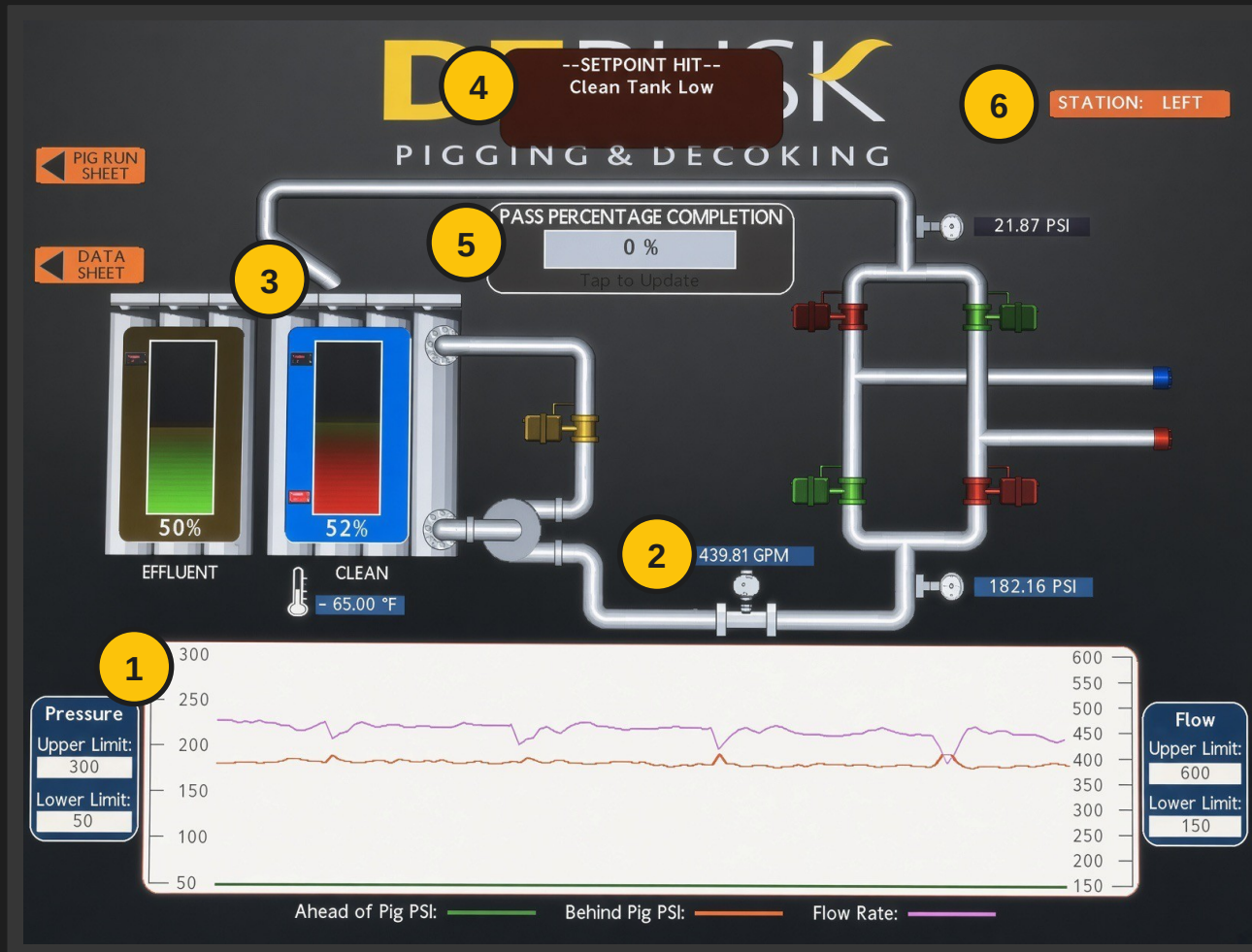
OPERATING PRESSURE, PSI

Normal pigging

Effective max



Real-time control at every operator station



- 1 Live trends**
Pressure ahead of and behind the pig, plus flow; coke shows as a spike
- 2 Flow & pressure**
GPM and psi at the pump, updated continuously
- 3 Tank levels**
Clean and effluent tanks on guided-radar indicators
- 4 Setpoint alarms**
Upper and lower limits on pressure and flow trigger alerts
- 5 Digital pig logs**
Pass completion, pig run sheet and data sheet recorded automatically
- 6 One station per pass**
Left, center and right stations run independently

California-ready, CARB-registered fleet

CARB Statewide Portable Equipment Registration (PERP); certificates shown

6

identical TriMax units, SCAQMD-ready

18

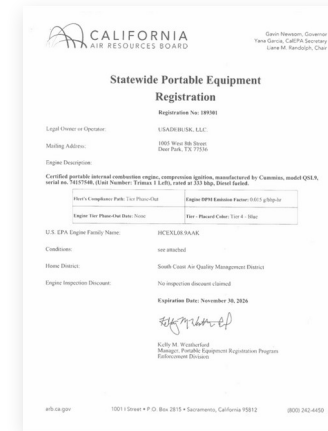
registered pumping engines

Tier 4

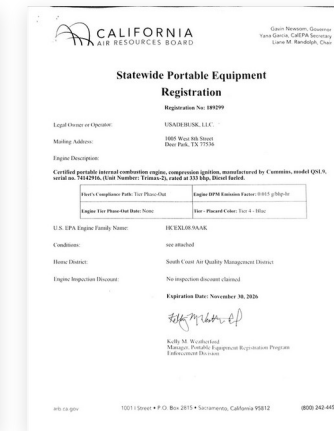
engines (blue placard)

0.015

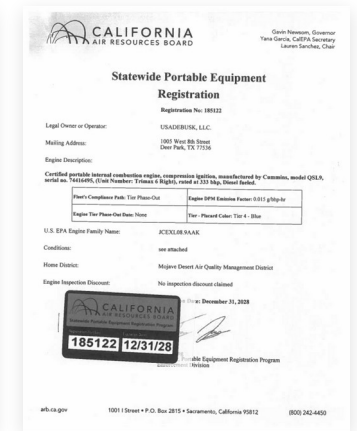
g/bhp-hr diesel PM emission factor



TriMax 1



TriMax 2



TriMax 6

CARB PERP registration certificates



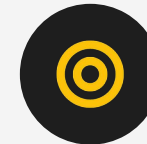
Support unit



300# launchers & receivers



Tested & certified
hard and flex pipe



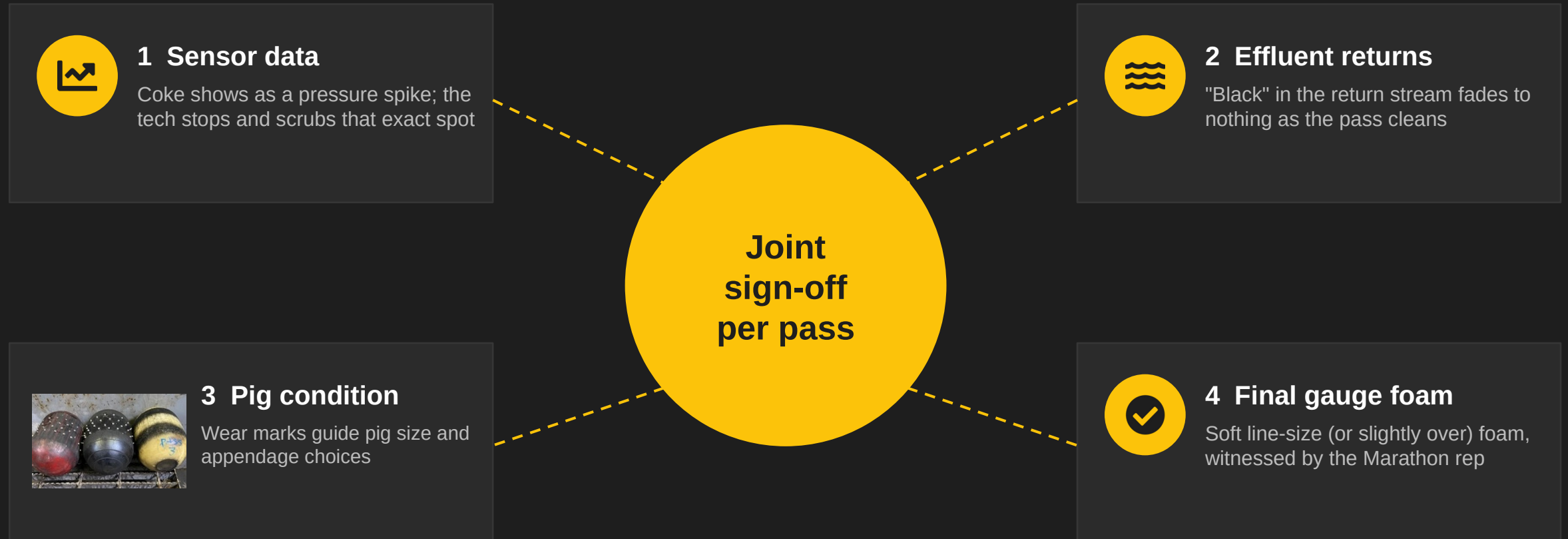
All pigs & foam
swabs



Optional: closed-loop
filtration

Verifying every pass is clean

Four independent indicators, reviewed side by side with Marathon



Proven in the field: successful sign-offs from all of our clients, and highly effective for first-attempt smart pig data.

QA/QC from walkdown to job book

Governed by the USA DeBusk QA/QC manual, submitted with the RFI response



Before

- Pre-job walkdown with operations
- Coil data checked against Marathon records
- JSA, permits and isolation confirmed
- Pig progression staged by heater



During

- Daily report to one point of contact
- Automated digital pig log per pass
- Flow tests before and after at the same rate
- JSA reviewed at every shift change



Close-out

- Per-pass cleanliness sign-off
- Final pig size recorded per pass
- Job book delivered to Marathon
- Post-job review and lessons learned

Ready when LAR needs us

Shops, equipment and crews in Southern California / LA Basin, with regional support

24/7

Dispatch and account manager, any hour

Same day

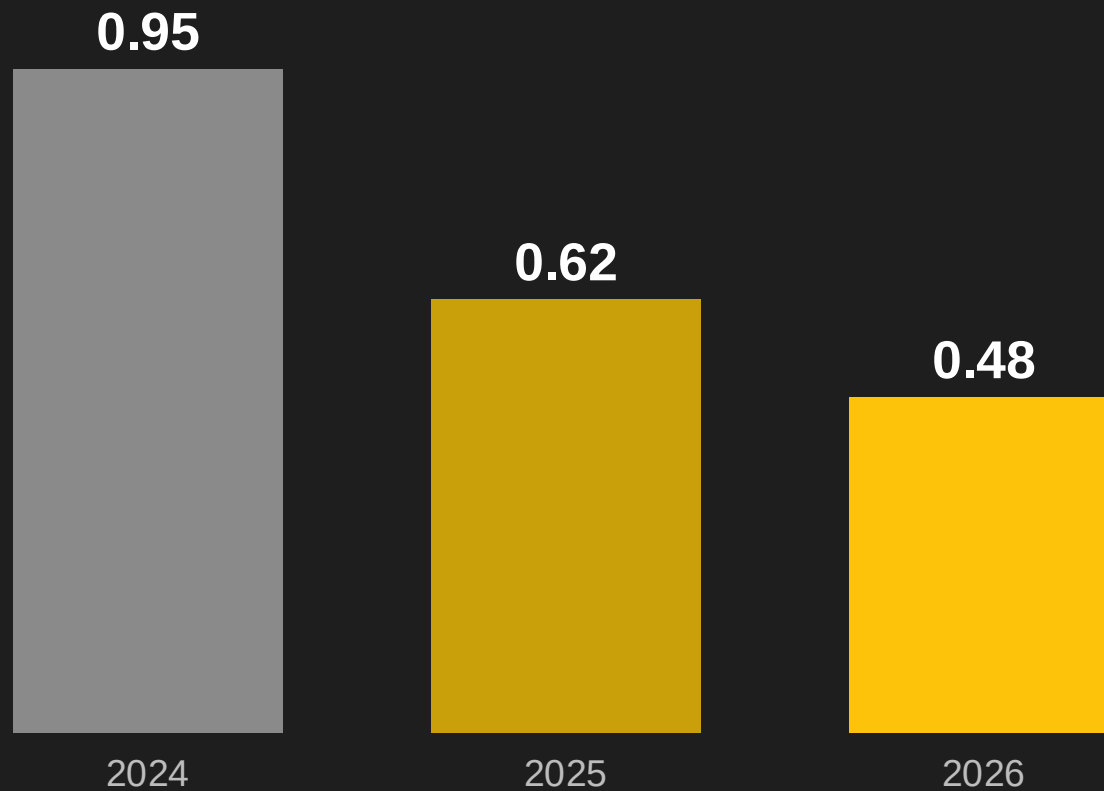
Response for local needs

< 24 hrs

Manpower & equipment anywhere else, from regional shops

Safety performance

Experience Modification Rate: 1.0 is the industry baseline; lower is better



A

ISNetworld safety grade

0.48

EMR, 2026

-49%

EMR since 2024

0

fatalities in 3 years

Risk mitigation in the field



JSA & permits

Job-specific JSA reviewed at every shift change; daily toolbox talks



Isolation first

Marathon's isolation and LOTO verified before any connection is made



Pressure control

Normal 150–300 psi; over-pressure checklist above 500 psi



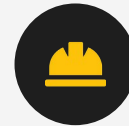
Line of fire

Bleed to zero at every pig change; nobody in line with lids or hoses



Hose integrity

Tested & certified hard and flex pipe; whip checks on every connection



PPE & coil care

FRC and face shields; no galvanized metal on stainless

Engineered for safety Retractable cam-actuated stairs, composite work platform, LED lighting



What sets USA DeBusk apart



Self-performed scope

Pigging, decoking and hydroblasting with our own equipment and SCAQMD rigs: one contractor, no subs



Largest standardized fleet

6 identical TriMax units, 18 engines, with a spare pump on board even on a 2-pass heater



Widest pig selection

Engineered decoking pigs matched to each coil and each coke condition

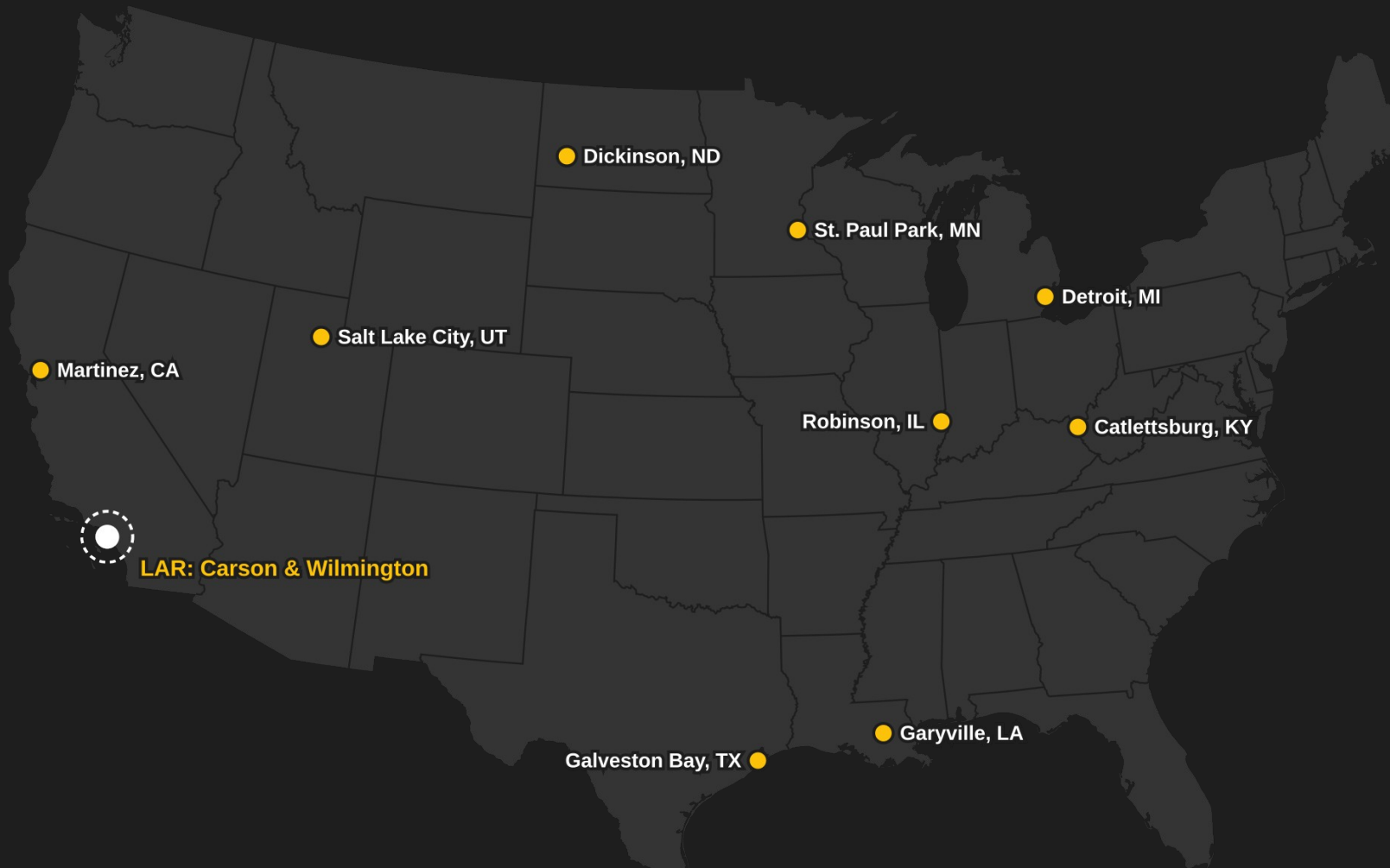


Data-driven, shared

Durations built from recorded actuals; lessons learned reviewed with Marathon after every job



Trusted across the MPC system



9

Marathon refineries with USA DeBusk references

- Named contacts at 8 sites (Galveston Bay on request), listed in the appendix
- Includes Martinez, CA, our West Coast MPC reference
- Crude, coker and vacuum heater experience

Coker heater case studies & lessons learned

Marathon Detroit, MI: 70H1 Coker

Decoked September 2024 and September 2026

6

passes, 3 cells

15,156'

coil footage

2

TriMax units

Scope All six passes, convection and radiant, pigged simultaneously on two TriMax units.

Result Two decokes on the same coker, two years apart: MPC brought us back.

CHS McPherson, KS: Coker HF-0012

6-pass coker heater, planned decoke, 2025

12,036'

coil footage

6

passes

85 h

pigging

Challenge Hard coke held pigs in the radiant section.

Lesson learned Stage heavy-duty scraper pigs for the radiant from day one, not as a mid-job change.

Why USA DeBusk for LAR



Proven experience

200+ years combined; 9 MPC references, including Detroit's coker



California-ready

CARB-registered Tier 4 fleet, Southern California footprint



Resilient execution

Triple-pass TriMax: a spare pump on every 2-pass heater



Verified clean

4-point verification and joint sign-off on every pass



Safe by the numbers

EMR 0.48, ISNetworld A, zero fatalities in 3 years





Thank you

Questions & discussion

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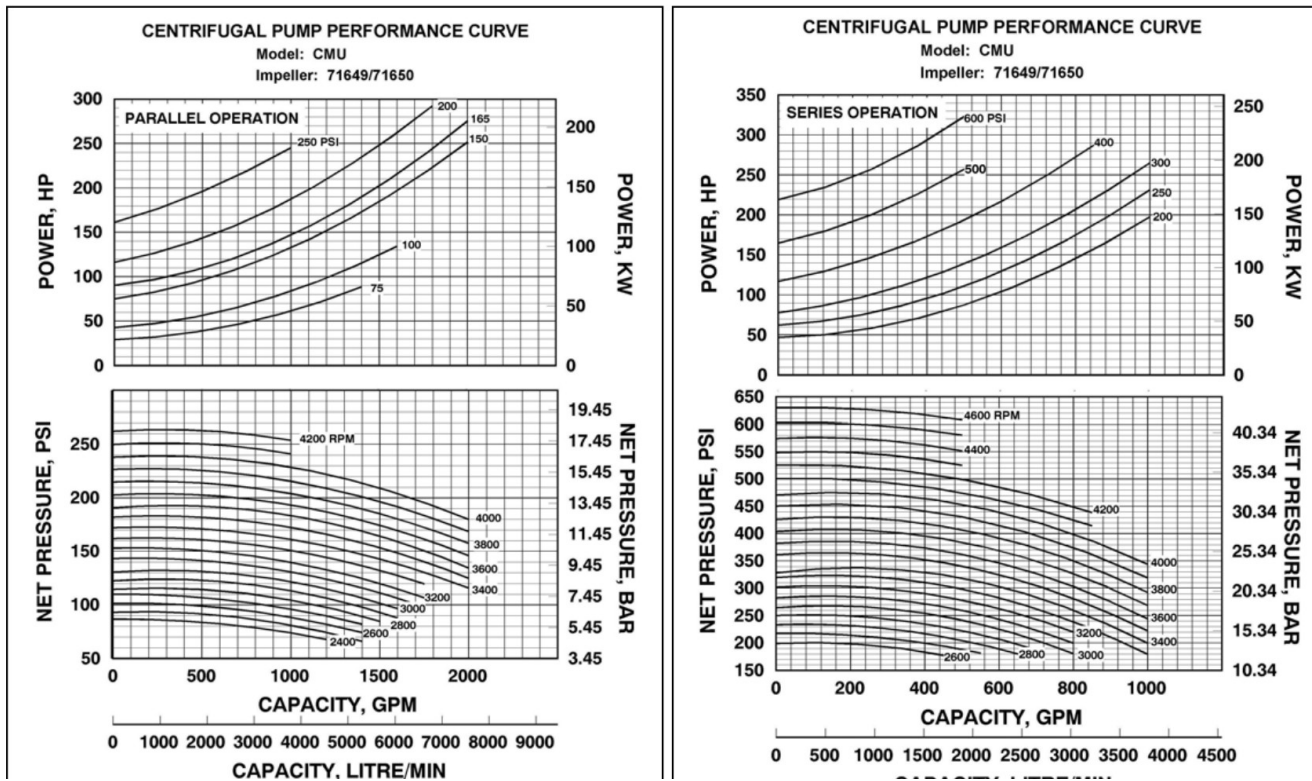
MPC references

Refinery	Contact	Role	Phone	Email
Catlettsburg, KY	Eddie Slone	T/A Planner	606-331-2585	slone@marathonpetroleum.com
	Ryan Rayburn	TAR / Construction Supervisor	606-331-0180	rdrayburn@marathonpetroleum.com
	Ben Holbrook	Turnaround Planning	606-369-6467	benlholbrook@marathonpetroleum.com
Salt Lake City, UT	Aaron Wiggins	Maintenance Supervisor / Turnarounds	801-703-6722	ATWiggins@marathonpetroleum.com
Martinez, CA	Kyle Yates	Project Engineer	925-316-9726	SYates2@marathonpetroleum.com
	Ryan Gorman		952-607-6853	RMGorman@marathonpetroleum.com
Dickinson, ND	Tyler Bauer	TA	701-516-3456	tbauer@marathonpetroleum.com
St. Paul Park, MN	Keith James	TA Planner	715-441-7986	TKJames@marathonpetroleum.com
Robinson, IL	Brian Maus	TA	618-544-2121	bkmaus@marathonpetroleum.com
Galveston Bay, TX	Available upon request			
Garyville, LA	Shawn Kramer	TA Planner	225-571-2689	skramer@marathonpetroleum.com
	Chad Childs	TA Supervisor	281-299-6088	cechilds@marathonpetroleum.com
Detroit, MI	Nate Lajiness	Turnaround Planner	313-297-6166	nlajiness@marathonpetroleum.com

TriMax pump performance

Waterous CMU two-stage pump, one per pass. Series operation is used for pigging.

Pump Performance Curves



150–300

psi normal pigging

500–550

psi effective max, after valve losses

600

psi rated pump maximum

Manufacturer curves: Waterous Company, form F-2692